15. Perform transformation using Direct Linear Transformation.

Aim:

To perform image transformation using the Direct Linear Transformation method with the OpenCV library.

Procedure:

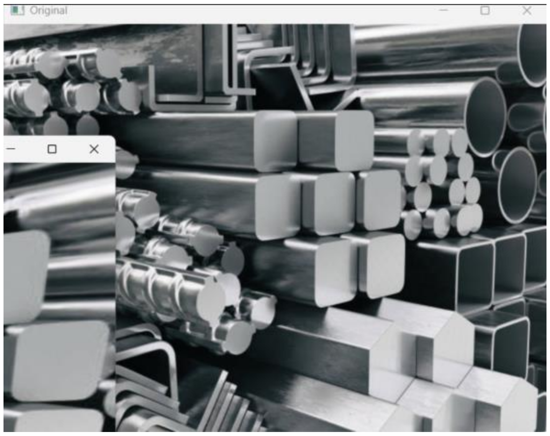
- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

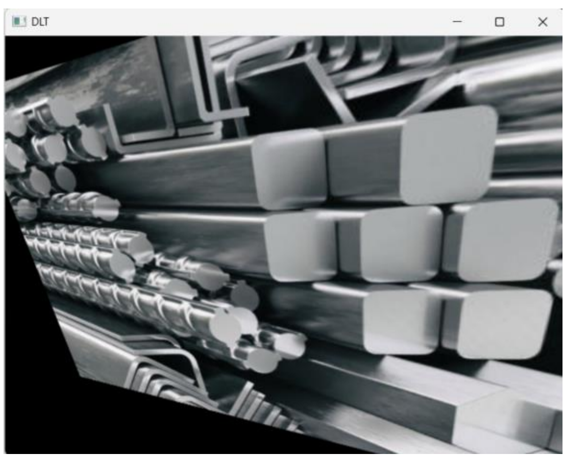
```
import cv2

import numpy as np



cap=cv2.VideoCapture(0)



while True:

    ret,frame=cap.read()

    if not ret:

        break



    rows,cols=frame.shape[:2]
```

```
    pts1=np.float32([[50,50],[cols-50,50],[50,rows-50],[cols-
50,rows-50]])
    pts2=np.float32([[0,100],[cols,0],[100,rows],[cols-100,rows]])

    M=cv2.getPerspectiveTransform(pts1,pts2)
    dst=cv2.warpPerspective(frame,M,(cols,rows))

    cv2.imshow("Original",frame)
    cv2.imshow("Perspective Video",dst)

    if cv2.waitKey(1)&0xFF==27:
        break

cap.release()
cv2.destroyAllWindows()
```

Sample Input



Sample Output



Result:

The input image was successfully transformed using the Direct Linear Transformation (DLT) method.